

RACHANA NIJALINGAPPA

AI Engineer | LLM & RAG Systems | Backend Engineering (Python, C#, .NET, Azure)

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linkedin.com/in/n-rachana | github.com/rachananijalingappa | [Portfolio](#) | Right to work in the UK

PROFESSIONAL SUMMARY

AI Engineer (MSc AI, Brunel) with 5.5 years of backend engineering in banking and NHS projects.

Builds agentic RAG apps in Python: LangGraph, FastAPI, ChromaDB, LLM evaluation and guardrails.

Brings banking experience in card tokenisation, test coverage (56% to 80%) and Azure to LLM delivery.

TECHNICAL SKILLS

Generative AI & LLMs: Retrieval-augmented generation (RAG), agentic workflows (LangGraph), LangChain, OpenAI API (GPT-4o-mini), prompt engineering, Cohere re-ranking, contextual compression, LLM-as-a-judge evaluation, prompt-injection guardrails

Vector Search: ChromaDB, embeddings, dense retrieval with re-ranking

Machine Learning: scikit-learn, LightGBM, Optuna, SHAP, TensorFlow/Keras (LSTM, 1D-CNN, autoencoders), SMOTE, Isolation Forest, PyTorch

Languages: Python, C#, SQL Server, JavaScript

Backend & APIs: FastAPI, Flask, .NET 8 / ASP.NET Core, async/await, REST APIs, microservices, JWT authentication, Entity Framework Core, CQRS/MediatR

Cloud & DevOps: Azure (Functions, Key Vault, Storage, Service Bus, API Management, Application Gateway, Application Insights, WebJobs), Docker, GitHub Actions CI/CD, Hugging Face Spaces

Testing & Observability: NUnit, Moq, pytest, LangSmith, Splunk, Observe

AI PROJECTS

Agentic RAG for Multi-Source Financial QA | MSc Dissertation

June 2026 - Sep 2026

- Built and compared five QA methods in LangGraph with GPT-4o-mini (no retrieval, dense RAG, HyDE, static source routing, adaptive self-correction) over 10-Q filings (3,641 chunks) and yfinance market data and news for five companies.
- Designed a 150-question benchmark (factual, numerical, sentiment, analytical, unanswerable) and evaluated 750 answers with correctness scoring, RAGAS, Precision@5 and MRR, using Wilson intervals, exact McNemar tests and paired bootstrap.
- Showed static routing raised correctness over single-source RAG from 56.7% to 91.8% on 97 scorable questions (34 gains, 0 losses), with 97.6% exact-match source selection; self-correction never beat it and roughly doubled latency.
- Classified failures with an eight-type taxonomy: confirmed errors came from numerical parsing (6) and retrieval of another company's figures (2); BERTScore missed the routing gain, and the automatic failure detector caught 1 of 8 failures.

Tech: Python, LangGraph, ChromaDB, GPT-4o-mini, RAGAS, BERTScore, yfinance

DocChat: Agentic RAG Microservice

May 2026 - Sep 2026

- Built a LangGraph routing workflow with GPT-4o-mini that sends each query to document retrieval or a web-search fallback, replacing a single-path RAG pipeline.
- Added contextual compression and Cohere re-ranking over ChromaDB retrieval to filter irrelevant context before generation.
- Wrote an evaluation script that generates test questions from document chunks and scores answers for faithfulness and relevance, plus an isolated LLM pre-check that screens inputs for prompt injection before retrieval.
- Deployed a FastAPI backend and Streamlit frontend in Docker on Hugging Face Spaces, with LangSmith tracing across the pipeline.

Tech: Python, LangGraph, LangChain, FastAPI, Streamlit, OpenAI API, Cohere API, ChromaDB, LangSmith, Docker

Credit Card Fraud Detection | ML Pipeline

Apr 2026

- Compared 13 models across classical ML, gradient boosting and deep learning (LSTM, 1D-CNN, autoencoder) on the Kaggle European credit card dataset (283,726 transactions after removing duplicates, 0.17% fraud).

- Applied SMOTE to the training set only, keeping the test set untouched, and tuned LightGBM with Optuna (30 trials, 3-fold stratified CV); the best model reached PR-AUC 0.817 and F1 0.859 (97.3% precision, 76.8% recall) on the test set.
- Used SHAP to identify the strongest fraud predictors and benchmarked unsupervised detection with Isolation Forest and K-Means.

Tech: Python, scikit-learn, LightGBM, Optuna, TensorFlow/Keras, SHAP, Pandas

NHS Appointment Optimisation | STAHN Hackathon (Brunel & Hillingdon Hospitals NHS FT) Feb 2026 - Mar 2026

- Built the working prototype in a 5-person team, creating a FastAPI and Streamlit tool to cut wasted outpatient appointments, designed against a target of reducing the did-not-attend rate from 8% to 2%.
- Generated synthetic data modelled on the provided hospital dataset after finding data quality issues in it.
- Built a rule-based scoring engine that matches waiting patients to cancelled slots and sends app-based offers within a 72-hour recovery window, replacing manual phone calls.
- Delivered a 13-metric KPI dashboard and a 3-tier reminder escalation flow across patient and admin views.

Tech: Python, FastAPI, Streamlit, synthetic data

PROFESSIONAL EXPERIENCE

Software Engineer | Commonwealth Bank of Australia | Bengaluru, India Dec 2023 - Jan 2025

- Built and secured card tokenisation APIs for tap-and-pay and other transactions, covering token issuance, lifecycle management and vault calls, on Azure Functions and Service Bus with Key Vault and Storage.
- Raised NUnit unit test coverage from 56% to 80% across the codebase, writing the tests single-handedly.
- Migrated 21 observability dashboards and their log streams from Splunk to Observe and walked teammates through the new setup.

Tech: .NET 8, C#, Azure (Functions, Storage, Key Vault, Service Bus), NUnit, Observe, Splunk, SQL Server

Associate Technology L2 (Software Engineer) | Publicis Sapient | Bengaluru, India May 2022 - Dec 2023

- As sole developer, rebuilt a legacy .NET Core workflow in Python ETL pipeline for a global asset management client, automating the daily collection and cleaning of 10-12 third-party financial data files and replacing 2-3 hours of manual work a day.
- Scheduled daily Azure WebJobs that load the data into CSV files and SQL tables, with retries, email alerts and logging on failure, and fed the cleaned data to Angular dashboards.
- Built GitHub Actions CI/CD pipelines for the data workflow.

Tech: Python (Flask), .NET 6, Azure (Functions, Service Bus, WebJobs), SQL Server, Angular, GitHub Actions, Docker, Pandas, NumPy

Software Engineer | Harman | Bengaluru, India Jul 2019 - May 2022

- Built and maintained the Angular UI for an NHS application during COVID-19, used by admin staff and nurses to book appointments and view patient demographics, records and files.
- Developed IoT applications for Legrand's connected lighting, secured with JWT authentication, Azure API Management and Application Gateway, with Azure Storage.
- Implemented event-driven processing with Azure Functions and Service Bus, maintaining 75% test coverage with NUnit and Moq.

Tech: .NET Core, C#, Angular, Azure (Functions, Service Bus, API Management, Application Gateway, Application Insights), SQL Server, NUnit, Moq

EDUCATION

MSc Artificial Intelligence | Brunel University London Sep 2025 - Sep 2026

- Modules: Deep Learning, NLP, Machine Learning, Computer Vision, AI Ethics, Data Visualisation.

BEng Electronics & Communication Engineering | MVJ College of Engineering, Bengaluru June 2015 – June 2019

- First Class with Distinction (CGPA 8.28/10).

AWARDS

- 2nd Prize, MSc category, INWED 2026 Inter-University Women in Engineering Poster Competition (ethical AI in healthcare).